



Gamma-ray transfer in Type Ia supernovae

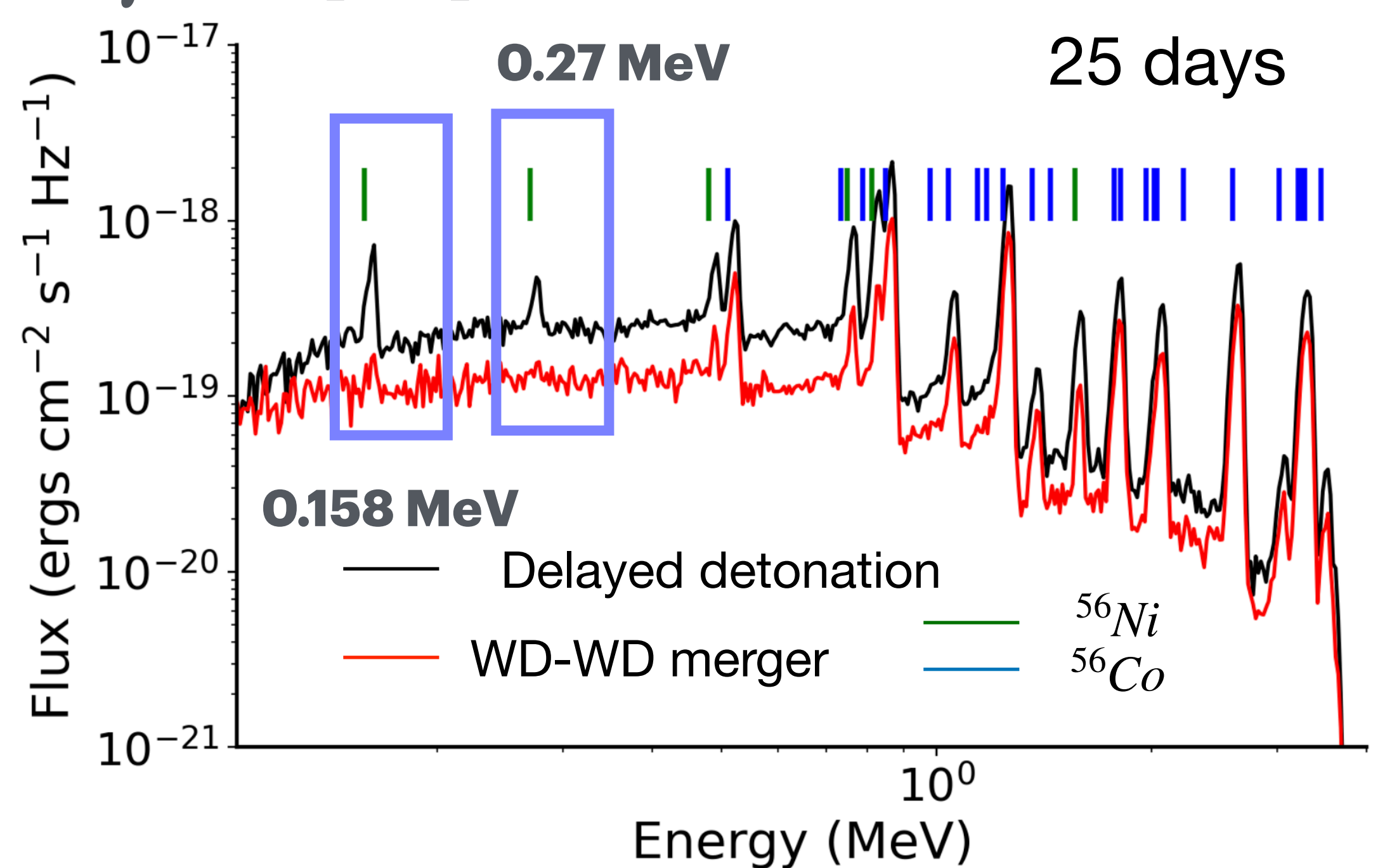
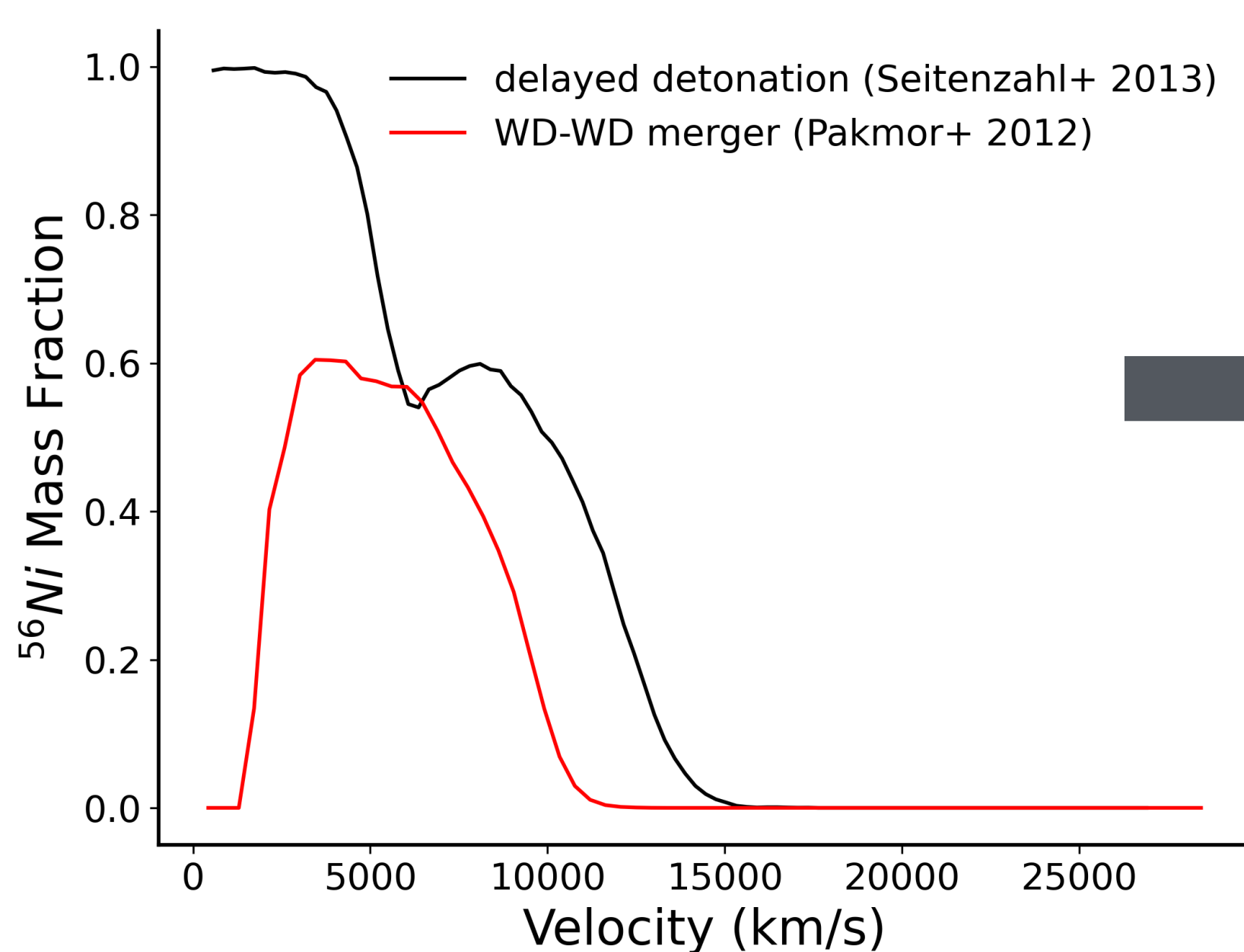
Anirban Dutta, Andrew Fullard, Wolfgang Kerzendorf (MSU), Jack O'Brien (UIUC)



Looking into the gamma-ray window to probe isotope mass and spatial distribution of isotopes coming from different Type Ia supernova models.

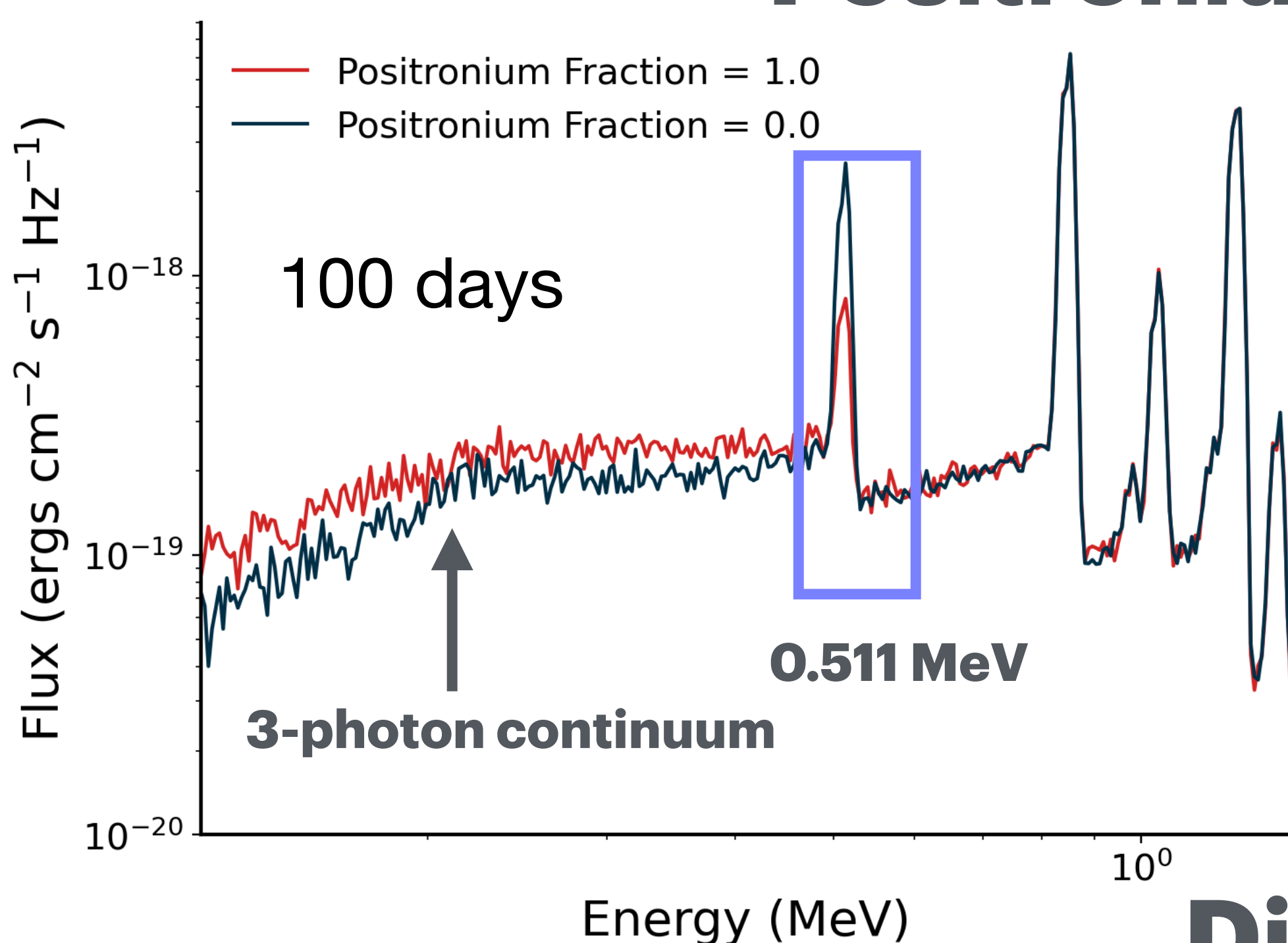
- Open Source γ -ray module to TARDIS (Kerzendorf and Sim 2014).
- Based on Monte Carlo methods.
- Photon interactions - photopair production, Compton scattering, photoelectric absorption.
- Positronium fraction.
- γ -ray spectra, light curves.
- Energy deposition by γ -rays and positrons.

Explosion model $\rightarrow$ γ -ray spectra



Comparison of the early time γ -ray spectra of DDT and merger models show the differences in the spectra of these models due to differences in the isotopic distribution.

Positronium contribution



- Positronium is a bound state of an e^- and e^+ .
- It annihilates to produce two 511 keV photons with probability of 25%.
- 75% of annihilations occur by producing 3-photons which shares the 1022 keV annihilation energy.
- The 3-photons form a continuum up to 511 keV. (Ore and Powell 1949)

Discussions

1. γ -ray spectra traces the spatial distribution and mass of the radioactive isotopes - potential diagnostic of progenitor properties.
2. Positronium formation changes the flux of the annihilation line (511 keV) and contributes to the continuum.